

ECEN 662: Estimation & Detection Theory

Lecture 16: Empirical Bayes Estimation

Tie Liu

Texas A&M University

Empirical Bayes

- In standard Bayes estimation, it is assumed that the unknown parameter Θ follows a *prior distribution* $\pi(\theta)$, and a *Bayes* estimator is one that minimizes the Bayesian risk

Empirical Bayes

- In standard Bayes estimation, it is assumed that the unknown parameter Θ follows a *prior distribution* $\pi(\theta)$, and a *Bayes* estimator is one that minimizes the Bayesian risk
- The prior distribution is chosen *before* any observation is made, so from the Bayesian point of view, this approach is *pure*

Empirical Bayes

- In standard Bayes estimation, it is assumed that the unknown parameter Θ follows a *prior distribution* $\pi(\theta)$, and a *Bayes* estimator is one that minimizes the Bayesian risk
- The prior distribution is chosen *before* any observation is made, so from the Bayesian point of view, this approach is *pure*
- Alternatively, one may consider choosing the prior distribution *after* an observation has been made

Empirical Bayes

- In standard Bayes estimation, it is assumed that the unknown parameter Θ follows a *prior distribution* $\pi(\theta)$, and a *Bayes* estimator is one that minimizes the Bayesian risk
- The prior distribution is chosen *before* any observation is made, so from the Bayesian point of view, this approach is *pure*
- Alternatively, one may consider choosing the prior distribution *after* an observation has been made
- In this case, one may attempt to build an estimator based on direct or indirect *learning* of the prior distribution from the observation

Empirical Bayes

- In standard Bayes estimation, it is assumed that the unknown parameter Θ follows a *prior distribution* $\pi(\theta)$, and a *Bayes* estimator is one that minimizes the Bayesian risk
- The prior distribution is chosen *before* any observation is made, so from the Bayesian point of view, this approach is *pure*
- Alternatively, one may consider choosing the prior distribution *after* an observation has been made
- In this case, one may attempt to build an estimator based on direct or indirect *learning* of the prior distribution from the observation
- When the learning of the prior distribution follows a *frequentist* approach, the overall approach for building such estimators is called *empirical Bayes*

Empirical Bayes

- In standard Bayes estimation, it is assumed that the unknown parameter Θ follows a *prior distribution* $\pi(\theta)$, and a *Bayes* estimator is one that minimizes the Bayesian risk
- The prior distribution is chosen *before* any observation is made, so from the Bayesian point of view, this approach is *pure*
- Alternatively, one may consider choosing the prior distribution *after* an observation has been made
- In this case, one may attempt to build an estimator based on direct or indirect *learning* of the prior distribution from the observation
- When the learning of the prior distribution follows a *frequentist* approach, the overall approach for building such estimators is called *empirical Bayes*
- Next, we shall illustrate how this approach works with an example that stunned the entire statistics community in 1961

Gaussian with unknown means

- Let $X = (X_1, \dots, X_n)$, where

$$X_i \stackrel{ind}{\sim} \mathcal{N}(\theta_i, 1)$$

Here, $\theta = (\theta_1, \dots, \theta_n)$ are the unknown parameters

Gaussian with unknown means

- Let $X = (X_1, \dots, X_n)$, where

$$X_i \stackrel{ind}{\sim} \mathcal{N}(\theta_i, 1)$$

Here, $\theta = (\theta_1, \dots, \theta_n)$ are the unknown parameters

- A *natural* estimator for θ_i is X_i , which is the MLE, the UMVUE, and a generalized Bayes estimator under the squared-error loss

Gaussian with unknown means

- Let $X = (X_1, \dots, X_n)$, where

$$X_i \stackrel{\text{ind}}{\sim} \mathcal{N}(\theta_i, 1)$$

Here, $\theta = (\theta_1, \dots, \theta_n)$ are the unknown parameters

- A *natural* estimator for θ_i is X_i , which is the MLE, the UMVUE, and a generalized Bayes estimator under the squared-error loss
- Now suppose we want to *simultaneously* estimate θ_i for all $i \in [n]$, and our loss function is the (completely natural) *total* squared-error loss

$$L(\theta, d) = \sum_{i=1}^n (\theta_i - d_i)^2$$

Gaussian with unknown means

- Let $X = (X_1, \dots, X_n)$, where

$$X_i \stackrel{\text{ind}}{\sim} \mathcal{N}(\theta_i, 1)$$

Here, $\theta = (\theta_1, \dots, \theta_n)$ are the unknown parameters

- A *natural* estimator for θ_i is X_i , which is the MLE, the UMVUE, and a generalized Bayes estimator under the squared-error loss
- Now suppose we want to *simultaneously* estimate θ_i for all $i \in [n]$, and our loss function is the (completely natural) *total* squared-error loss

$$L(\theta, d) = \sum_{i=1}^n (\theta_i - d_i)^2$$

- It is only natural to think that $X = (X_1, \dots, X_n)$ is (at least) admissible. A stunning result of James and Stein in 1961, however, showed that this is actually *not* the case when $n \geq 3$

James-Stein estimator

- In 1961, James and Stein introduced what is now known as the *James-Stein estimator*:

$$\delta_i(X) = \left(1 - \frac{n-2}{\|X\|^2}\right) X_i$$

James-Stein estimator

- In 1961, James and Stein introduced what is now known as the *James-Stein estimator*:

$$\delta_i(X) = \left(1 - \frac{n-2}{\|X\|^2}\right) X_i$$

- To compare the risks of the natural estimator X and the James-Stein estimator $\delta(X)$, note that

$$\begin{aligned}\|\delta(X) - \theta\|^2 &= \|(\delta(X) - X) + (X - \theta)\|^2 \\ &= \|\delta(X) - X\|^2 + \|X - \theta\|^2 + 2(\delta(X) - X)^t(X - \theta) \\ &= \frac{(n-2)^2}{\|X\|^2} + \|X - \theta\|^2 - 2(n-2) \left(\frac{X}{\|X\|^2}\right)^t (X - \theta)\end{aligned}$$

James-Stein estimator

- In 1961, James and Stein introduced what is now known as the *James-Stein estimator*:

$$\delta_i(X) = \left(1 - \frac{n-2}{\|X\|^2}\right) X_i$$

- To compare the risks of the natural estimator X and the James-Stein estimator $\delta(X)$, note that

$$\begin{aligned}\|\delta(X) - \theta\|^2 &= \|(\delta(X) - X) + (X - \theta)\|^2 \\ &= \|\delta(X) - X\|^2 + \|X - \theta\|^2 + 2(\delta(X) - X)^t(X - \theta) \\ &= \frac{(n-2)^2}{\|X\|^2} + \|X - \theta\|^2 - 2(n-2) \left(\frac{X}{\|X\|^2}\right)^t (X - \theta)\end{aligned}$$

- Taking expectations on both sides gives

$$\begin{aligned}R_\delta(\theta) &= R_X(\theta) + (n-2)^2 \mathbb{E}_\theta \left[\frac{1}{\|X\|^2} \right] - \\ &\quad 2(n-2) \mathbb{E}_\theta \left[\left(\frac{X}{\|X\|^2}\right)^t (X - \theta) \right]\end{aligned}$$

- (*Stein's lemma*) Let $Z = (Z_1, \dots, Z_n)$ be a *standard* Gaussian vector, and let $g : \mathcal{R}^n \rightarrow \mathcal{R}^n$ be a differential function such that

$$\mathbb{E} \left[\sqrt{\sum_{i=1}^n \sum_{j=1}^n \left(\frac{\partial}{\partial Z_j} g_i(Z) \right)^2} \right] < \infty$$

Then

$$\mathbb{E}[g(Z)^t Z] = \sum_{i=1}^n \mathbb{E} \left[\frac{\partial}{\partial Z_i} g_i(Z) \right]$$

- (*Stein's lemma*) Let $Z = (Z_1, \dots, Z_n)$ be a *standard* Gaussian vector, and let $g : \mathcal{R}^n \rightarrow \mathcal{R}^n$ be a differential function such that

$$\mathbb{E} \left[\sqrt{\sum_{i=1}^n \sum_{j=1}^n \left(\frac{\partial}{\partial Z_j} g_i(Z) \right)^2} \right] < \infty$$

Then

$$\mathbb{E}[g(Z)^t Z] = \sum_{i=1}^n \mathbb{E} \left[\frac{\partial}{\partial Z_i} g_i(Z) \right]$$

- By Stein's lemma,

$$\mathbb{E}_\theta \left[\left(\frac{X}{\|X\|^2} \right)^t (X - \theta) \right] = (n-2) \mathbb{E}_\theta \left[\frac{1}{\|X\|^2} \right]$$

- (*Stein's lemma*) Let $Z = (Z_1, \dots, Z_n)$ be a *standard* Gaussian vector, and let $g : \mathcal{R}^n \rightarrow \mathcal{R}^n$ be a differential function such that

$$\mathbb{E} \left[\sqrt{\sum_{i=1}^n \sum_{j=1}^n \left(\frac{\partial}{\partial Z_j} g_i(Z) \right)^2} \right] < \infty$$

Then

$$\mathbb{E}[g(Z)^t Z] = \sum_{i=1}^n \mathbb{E} \left[\frac{\partial}{\partial Z_i} g_i(Z) \right]$$

- By Stein's lemma,

$$\mathbb{E}_\theta \left[\left(\frac{X}{\|X\|^2} \right)^t (X - \theta) \right] = (n - 2) \mathbb{E}_\theta \left[\frac{1}{\|X\|^2} \right]$$

- It thus follows that

$$R_\delta(\theta) = R_X(\theta) - (n - 2)^2 \mathbb{E}_\theta \left[\frac{1}{\|X\|^2} \right] < R_X(\theta), \quad \forall n \geq 3$$

James-Stein as a shrinkage estimator

- James-Stein estimator is a wonderful *precise* result

James-Stein as a shrinkage estimator

- James-Stein estimator is a wonderful *precise* result
- The most striking thing about the James-Stein estimator is that the estimate of each θ_i depends on the *entire* observation, even though *only* the distribution of X_i depends on θ_i

James-Stein as a shrinkage estimator

- James-Stein estimator is a wonderful *precise* result
- The most striking thing about the James-Stein estimator is that the estimate of each θ_i depends on the *entire* observation, even though *only* the distribution of X_i depends on θ_i
- The main intuition for the James-Stein estimator is that

$$R_X(\theta) = \mathbb{E}_\theta[\|X\|^2] - \|\theta\|^2$$

However,

$$\mathbb{E}_\theta[\|X\|^2] = n + \|\theta\|^2 \gg \|\theta\|^2$$

especially when n is large

James-Stein as a shrinkage estimator

- James-Stein estimator is a wonderful *precise* result
- The most striking thing about the James-Stein estimator is that the estimate of each θ_i depends on the *entire* observation, even though *only* the distribution of X_i depends on θ_i
- The main intuition for the James-Stein estimator is that

$$R_X(\theta) = \mathbb{E}_\theta[\|X\|^2] - \|\theta\|^2$$

However,

$$\mathbb{E}_\theta[\|X\|^2] = n + \|\theta\|^2 \gg \|\theta\|^2$$

especially when n is large

- It is thus natural to consider *shrinking* the natural estimator X to make up the gap between $\mathbb{E}_\theta[\|X\|^2]$ and $\|\theta\|^2$

James-Stein as a shrinkage estimator

- James-Stein estimator is a wonderful *precise* result
- The most striking thing about the James-Stein estimator is that the estimate of each θ_i depends on the *entire* observation, even though *only* the distribution of X_i depends on θ_i
- The main intuition for the James-Stein estimator is that

$$R_X(\theta) = \mathbb{E}_\theta[\|X\|^2] - \|\theta\|^2$$

However,

$$\mathbb{E}_\theta[\|X\|^2] = n + \|\theta\|^2 \gg \|\theta\|^2$$

especially when n is large

- It is thus natural to consider *shrinking* the natural estimator X to make up the gap between $\mathbb{E}_\theta[\|X\|^2]$ and $\|\theta\|^2$
- While *shrinkage* is a common characteristic of *Bayes* estimators, here that the shrinkage factor $1 - (n - 2)/\|X\|^2$ depends on the observation X naturally reminds us of *empirical* Bayes

James-Stein as an empirical Bayes estimator

- To see how the James-Stein estimator can be derived via *empirical* Bayes estimation, consider the family of priors

$$\Theta_i \stackrel{iid}{\sim} \mathcal{N}(0, A)$$

parameterized by $A > 0$

James-Stein as an empirical Bayes estimator

- To see how the James-Stein estimator can be derived via *empirical* Bayes estimation, consider the family of priors

$$\Theta_i \stackrel{iid}{\sim} \mathcal{N}(0, A)$$

parameterized by $A > 0$

- The *Bayes* estimator with respect to the squared-error loss is

$$\delta_{\pi_A} = \left(1 - \frac{1}{A+1}\right) X$$

James-Stein as an empirical Bayes estimator

- To see how the James-Stein estimator can be derived via *empirical* Bayes estimation, consider the family of priors

$$\Theta_i \stackrel{iid}{\sim} \mathcal{N}(0, A)$$

parameterized by $A > 0$

- The *Bayes* estimator with respect to the squared-error loss is

$$\delta_{\pi_A} = \left(1 - \frac{1}{A+1}\right) X$$

- To find an estimate of $\frac{1}{A+1}$ based on the observation A , note that

$$X_i \stackrel{iid}{\sim} \mathcal{N}(0, A+1)$$

- One possible choice is to find the *UMVUE* for $\frac{1}{A+1}$

- To do so, simply note that $\|X\|^2$ is a complete sufficient statistic for A , and

$$\mathbb{E}_A \left[\frac{1}{\|X\|^2} \right] = \frac{1}{(n-2)(A+1)}$$

- To do so, simply note that $\|X\|^2$ is a complete sufficient statistic for A , and

$$\mathbb{E}_A \left[\frac{1}{\|X\|^2} \right] = \frac{1}{(n-2)(A+1)}$$

- It follows immediately that the MVUE of $\frac{1}{A+1}$ is

$$\frac{n-2}{\|X\|^2}$$

- To do so, simply note that $\|X\|^2$ is a complete sufficient statistic for A , and

$$\mathbb{E}_A \left[\frac{1}{\|X\|^2} \right] = \frac{1}{(n-2)(A+1)}$$

- It follows immediately that the MVUE of $\frac{1}{A+1}$ is

$$\frac{n-2}{\|X\|^2}$$

- Substituting

$$\frac{1}{A+1} \approx \frac{n-2}{\|X\|^2}$$

into the Bayes estimator

$$\delta_{\pi_A} = \left(1 - \frac{1}{A+1} \right) X$$

immediately gives rise to the James-Stein estimator

Shrinkage to the mean

- Assume that

$$X_i \stackrel{iid}{\sim} \mathcal{N}(\theta_i, \sigma_0^2)$$

where σ_0^2 is known. For any $i \in [n]$, let

$$\hat{\theta}_i = \bar{X} + \left(1 - \frac{(n-3)\sigma_0^2}{S}\right) (X_i - \bar{X})$$

where

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$S = \sum_{i=1}^n (X_i - \bar{X})^2$$

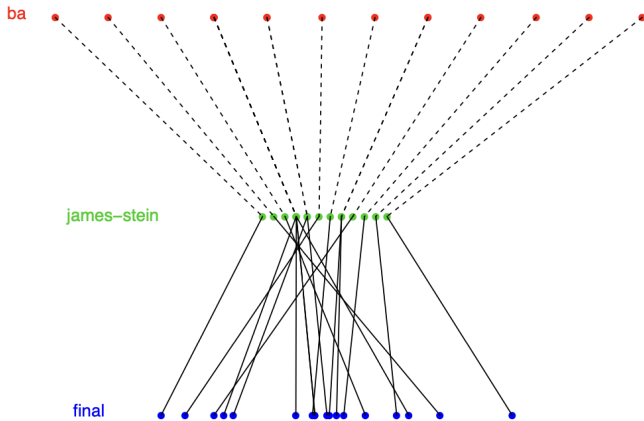
For any $n \geq 4$, we have

$$\mathbb{E}_{\theta}[\|\hat{\theta} - \theta\|^2] < n\sigma_0^2$$

James-Stein estimator in the wild

Table 6: Batting averages of 18 baseball players. *Column 1*, observed batting average after 45 attempts. *Column 2*, James–Stein estimate (3.4), $\sigma_0^2 = 0.066$. *Column 3*, average for remainder of the 1970 season. *Column 4*, number of attempts in remainder of the season.

	MLE	JS	final avg	# final AB
Clemente	.400	.294	.346	367
F. Robinson	.378	.289	.298	426
F. Howard	.356	.284	.276	521
Johnstone	.333	.279	.222	275
Berry	.311	.275	.273	418
Spencer	.311	.275	.270	466
Kessinger	.289	.270	.263	586
L. Alvarado	.267	.265	.210	138
Santo	.244	.261	.269	510
Swoboda	.244	.261	.230	200
Unser	.222	.256	.264	277
Williams	.222	.256	.256	270
Scott	.222	.256	.303	435
Petrocelli	.222	.256	.264	538
E. Rodriguez	.222	.256	.226	186
Campaneris	.200	.251	.285	558
Munson	.178	.246	.316	408
Alvis	.156	.242	.200	70



- The prediction error for the MLE and the James-Stein estimator are:

$$\|\hat{\theta}_{MLE}(x_{current}) - x_{final}\|^2 = 0.0745$$

$$\|\hat{\theta}_{JS}(x_{current}) - x_{final}\|^2 = 0.0213$$

- The prediction error for the MLE and the James-Stein estimator are:

$$\|\hat{\theta}_{MLE}(x_{current}) - x_{final}\|^2 = 0.0745$$

$$\|\hat{\theta}_{JS}(x_{current}) - x_{final}\|^2 = 0.0213$$

- Despite that the likelihood is *not* exactly Gaussian, the James-Stein rule reduces prediction error by a factor of more than 3

- The prediction error for the MLE and the James-Stein estimator are:

$$\|\hat{\theta}_{MLE}(x_{current}) - x_{final}\|^2 = 0.0745$$

$$\|\hat{\theta}_{JS}(x_{current}) - x_{final}\|^2 = 0.0213$$

- Despite that the likelihood is *not* exactly Gaussian, the James-Stein rule reduces prediction error by a factor of more than 3
- Baseball fans know that Clemente was the best hitter of his generation. Putting him in a James-Stein analysis with 17 less accomplished hitters almost guarantees an underestimate of his true ability

- The prediction error for the MLE and the James-Stein estimator are:

$$\|\hat{\theta}_{MLE}(x_{current}) - x_{final}\|^2 = 0.0745$$

$$\|\hat{\theta}_{JS}(x_{current}) - x_{final}\|^2 = 0.0213$$

- Despite that the likelihood is *not* exactly Gaussian, the James-Stein rule reduces prediction error by a factor of more than 3
- Baseball fans know that Clemente was the best hitter of his generation. Putting him in a James-Stein analysis with 17 less accomplished hitters almost guarantees an underestimate of his true ability
- James-Stein shrinkage is a winning tactic for minimizing total squared-error loss, but risks “sacrificing the extremes for the sake of the center”

- The prediction error for the MLE and the James-Stein estimator are:

$$\begin{aligned}\|\hat{\theta}_{MLE}(x_{current}) - x_{final}\|^2 &= 0.0745 \\ \|\hat{\theta}_{JS}(x_{current}) - x_{final}\|^2 &= 0.0213\end{aligned}$$

- Despite that the likelihood is *not* exactly Gaussian, the James–Stein rule reduces prediction error by a factor of more than 3
- Baseball fans know that Clemente was the best hitter of his generation. Putting him in a James–Stein analysis with 17 less accomplished hitters almost guarantees an underestimate of his true ability
- James–Stein shrinkage is a winning tactic for minimizing total squared-error loss, but risks “sacrificing the extremes for the sake of the center”
- More generally, empirical Bayes methods share data across individuals, for the benefit of overall performance but at possible peril to unusual individuals

From prior modeling to density estimation

- Previously we derived the James-Stein estimator by placing a *Gaussian* model on the prior distribution $\pi(\theta)$

From prior modeling to density estimation

- Previously we derived the James-Stein estimator by placing a *Gaussian* model on the prior distribution $\pi(\theta)$
- What if a Gaussian prior is *not* a good match to the problem at hand? Is there a version of the James–Stein rule applicable more *generally*?

From prior modeling to density estimation

- Previously we derived the James-Stein estimator by placing a *Gaussian* model on the prior distribution $\pi(\theta)$
- What if a Gaussian prior is *not* a good match to the problem at hand? Is there a version of the James–Stein rule applicable more *generally*?
- Below we shall provide a positive answer that is based on *Tweedie's formula*

From prior modeling to density estimation

- Previously we derived the James-Stein estimator by placing a *Gaussian* model on the prior distribution $\pi(\theta)$
- What if a Gaussian prior is *not* a good match to the problem at hand? Is there a version of the James–Stein rule applicable more *generally*?
- Below we shall provide a positive answer that is based on *Tweedie's formula*
- Consider the following Bayesian model underlying our empirical Bayes estimate:

$$\Theta \sim \pi(\theta), \quad \text{and} \quad X|\Theta = \theta \sim \mathcal{N}(\theta, 1)$$

From prior modeling to density estimation

- Previously we derived the James-Stein estimator by placing a *Gaussian* model on the prior distribution $\pi(\theta)$
- What if a Gaussian prior is *not* a good match to the problem at hand? Is there a version of the James-Stein rule applicable more *generally*?
- Below we shall provide a positive answer that is based on *Tweedie's formula*
- Consider the following Bayesian model underlying our empirical Bayes estimate:

$$\Theta \sim \pi(\theta), \quad \text{and} \quad X|\Theta = \theta \sim \mathcal{N}(\theta, 1)$$

- Let $m_\pi(x)$ be the marginal density of X . Then

$$m_\pi(x) = \int_{\Omega} \varphi(x - \theta) \pi(\theta) d\theta$$

where the φ is the density function of a standard Gaussian variable

- (*Tweedie's formula*) The posterior mean estimator

$$\delta_{\pi}(x) = \mathbb{E}[\Theta|X = x]$$

can be conveniently expressed in terms of the marginal density $m_{\pi}(x)$ as follows:

$$\delta_{\pi}(x) = x + \frac{d}{dx} \log m_{\pi}(x)$$

- (*Tweedie's formula*) The posterior mean estimator

$$\delta_{\pi}(x) = \mathbb{E}[\Theta|X = x]$$

can be conveniently expressed in terms of the marginal density $m_{\pi}(x)$ as follows:

$$\delta_{\pi}(x) = x + \frac{d}{dx} \log m_{\pi}(x)$$

- Tweedie's formula immediately suggests the following *empirical Bayes* estimation procedure:

- (*Tweedie's formula*) The posterior mean estimator

$$\delta_\pi(x) = \mathbb{E}[\Theta|X = x]$$

can be conveniently expressed in terms of the marginal density $m_\pi(x)$ as follows:

$$\delta_\pi(x) = x + \frac{d}{dx} \log m_\pi(x)$$

- Tweedie's formula immediately suggests the following *empirical Bayes* estimation procedure:
 - Estimate the marginal density $m_\pi(x)$ by a *smooth* curve $\hat{m}_\pi(x)$ drawn through the bar tops of the histogram for $x_1, \dots, x_n$

- (*Tweedie's formula*) The posterior mean estimator

$$\delta_\pi(x) = \mathbb{E}[\Theta|X = x]$$

can be conveniently expressed in terms of the marginal density $m_\pi(x)$ as follows:

$$\delta_\pi(x) = x + \frac{d}{dx} \log m_\pi(x)$$

- Tweedie's formula immediately suggests the following *empirical Bayes* estimation procedure:
 - Estimate the marginal density $m_\pi(x)$ by a *smooth* curve $\hat{m}_\pi(x)$ drawn through the bar tops of the histogram for $x_1, \dots, x_n$
 - Estimate the posterior mean as

$$\hat{\delta}_\pi(x) = x + \frac{d}{dx} \log \hat{m}_\pi(x)$$

- (*Tweedie's formula*) The posterior mean estimator

$$\delta_\pi(x) = \mathbb{E}[\Theta|X = x]$$

can be conveniently expressed in terms of the marginal density $m_\pi(x)$ as follows:

$$\delta_\pi(x) = x + \frac{d}{dx} \log m_\pi(x)$$

- Tweedie's formula immediately suggests the following *empirical Bayes* estimation procedure:
 - Estimate the marginal density $m_\pi(x)$ by a *smooth* curve $\hat{m}_\pi(x)$ drawn through the bar tops of the histogram for $x_1, \dots, x_n$
 - Estimate the posterior mean as

$$\hat{\delta}_\pi(x) = x + \frac{d}{dx} \log \hat{m}_\pi(x)$$

- Note that in the above procedure, a model is placed on the marginal density $m_\pi(x)$, instead of the prior distribution $\pi(\theta)$

The missing species problem

- *Corbet*, a leading naturalist, had been trapping butterflies in Malaysia (then Malaya) for two years:

Table 2: Butterfly data; number y_x of species seen x times each in two years of trapping; 118 species trapped just once, 74 trapped twice each, etc.

x	1	2	3	4	5	6	7	8	9	10	11	12
y_x	118	74	44	24	29	22	20	19	20	15	12	14
x	13	14	15	16	17	18	19	20	21	22	23	24
y_x	6	12	6	9	9	6	10	10	11	5	3	3

The missing species problem

- *Corbet*, a leading naturalist, had been trapping butterflies in Malaysia (then Malaya) for two years:

Table 2: Butterfly data; number y_x of species seen x times each in two years of trapping; 118 species trapped just once, 74 trapped twice each, etc.

x	1	2	3	4	5	6	7	8	9	10	11	12
y_x	118	74	44	24	29	22	20	19	20	15	12	14
x	13	14	15	16	17	18	19	20	21	22	23	24
y_x	6	12	6	9	9	6	10	10	11	5	3	3

- Note that in the table there is *no* $x = 0$ entry: These are the “missing species”

The missing species problem

- *Corbet*, a leading naturalist, had been trapping butterflies in Malaysia (then Malaya) for two years:

Table 2: Butterfly data; number y_x of species seen x times each in two years of trapping; 118 species trapped just once, 74 trapped twice each, etc.

x	1	2	3	4	5	6	7	8	9	10	11	12
y_x	118	74	44	24	29	22	20	19	20	15	12	14
x	13	14	15	16	17	18	19	20	21	22	23	24
y_x	6	12	6	9	9	6	10	10	11	5	3	3

- Note that in the table there is *no* $x = 0$ entry: These are the “missing species”
- Corbet asked *Fisher* a seemingly unanswerable missing-species question: If he continued trapping for one additional year, how many *new* species could he expect to see?

A Poisson process model

- Let X_i be the number of times species $i \in [S]$ is trapped in the original one unit of time (the unit being two years for Corbet), and let $X_i(t)$ the number of times trapped in a hypothetical future period of t units

A Poisson process model

- Let X_i be the number of times species $i \in [S]$ is trapped in the original one unit of time (the unit being two years for Corbet), and let $X_i(t)$ the number of times trapped in a hypothetical future period of t units
- Under the *Poisson process* assumption, we have

$$X_i \stackrel{ind}{\sim} \text{Poisson}(\theta_i)$$

$$X_i(t) \stackrel{ind}{\sim} \text{Poisson}(\theta_i t)$$

where $(X_i : i \in [S])$ and $(X_i(t) : i \in [S])$ are independent of each other

A Poisson process model

- Let X_i be the number of times species $i \in [S]$ is trapped in the original one unit of time (the unit being two years for Corbet), and let $X_i(t)$ the number of times trapped in a hypothetical future period of t units
- Under the *Poisson process* assumption, we have

$$X_i \stackrel{\text{ind}}{\sim} \text{Poisson}(\theta_i)$$

$$X_i(t) \stackrel{\text{ind}}{\sim} \text{Poisson}(\theta_i t)$$

where $(X_i : i \in [S])$ and $(X_i(t) : i \in [S])$ are independent of each other

- Under this model, the probability that species i is not seen in the initial trapping period but is seen in the future period is given by

$$\mathbb{P}_{\theta_i}(X_i = 0, X_i(t) > 0) = e^{-\theta_i}(1 - e^{-\theta_i t})$$

Constructing an empirical Bayes estimator

- Assuming that $\Theta_i \stackrel{iid}{\sim} \pi(\theta)$, the expected number of Corbet's hoped-for “new” species N is given by

$$\mathbb{E}[N] = S \int_0^\infty e^{-\theta}(1 - e^{-\theta t})\pi(\theta)d\theta$$

Constructing an empirical Bayes estimator

- Assuming that $\Theta_i \stackrel{iid}{\sim} \pi(\theta)$, the expected number of Corbet's hoped-for “new” species N is given by

$$\mathbb{E}[N] = S \int_0^\infty e^{-\theta} (1 - e^{-\theta t}) \pi(\theta) d\theta$$

- To construct an *empirical* Bayes estimator, expanding $1 - e^{-\theta t}$ in a Taylor series gives

$$\mathbb{E}[N] = S \int_0^\infty e^{-\theta} \left(\theta t - \frac{(\theta t)^2}{2!} + \frac{(\theta t)^3}{3!} - \dots \right) \pi(\theta) d\theta$$

Constructing an empirical Bayes estimator

- Assuming that $\Theta_i \stackrel{iid}{\sim} \pi(\theta)$, the expected number of Corbet's hoped-for “new” species N is given by

$$\mathbb{E}[N] = S \int_0^\infty e^{-\theta} (1 - e^{-\theta t}) \pi(\theta) d\theta$$

- To construct an *empirical* Bayes estimator, expanding $1 - e^{-\theta t}$ in a Taylor series gives

$$\mathbb{E}[N] = S \int_0^\infty e^{-\theta} \left(\theta t - \frac{(\theta t)^2}{2!} + \frac{(\theta t)^3}{3!} - \dots \right) \pi(\theta) d\theta$$

- Further note that for any $x \geq 1$,

$$\mathbb{E}[Y_x] = S \int_0^\infty \frac{e^{-\theta} \theta^x}{x!} \pi(\theta) d\theta$$

Constructing an empirical Bayes estimator

- Assuming that $\Theta_i \stackrel{iid}{\sim} \pi(\theta)$, the expected number of Corbet's hoped-for “new” species N is given by

$$\mathbb{E}[N] = S \int_0^\infty e^{-\theta} (1 - e^{-\theta t}) \pi(\theta) d\theta$$

- To construct an *empirical* Bayes estimator, expanding $1 - e^{-\theta t}$ in a Taylor series gives

$$\mathbb{E}[N] = S \int_0^\infty e^{-\theta} \left(\theta t - \frac{(\theta t)^2}{2!} + \frac{(\theta t)^3}{3!} - \dots \right) \pi(\theta) d\theta$$

- Further note that for any $x \geq 1$,

$$\mathbb{E}[Y_x] = S \int_0^\infty \frac{e^{-\theta} \theta^x}{x!} \pi(\theta) d\theta$$

- It follows immediately that

$$\mathbb{E}[N] = \mathbb{E}[Y_1]t - \mathbb{E}[Y_2]t^2 + \mathbb{E}[Y_3]t^3 - \dots$$

- Replacing $\mathbb{E}[Y_x]$ by the unbiased estimate y_x leads to *Good* and *Toulmin*'s intriguing estimator:

$$\widehat{\mathbb{E}[N]} = y_1 t - y_2 t^2 + y_3 t^3 - \dots$$

which played a significant role in their World War II code-breaking efforts

- Replacing $\mathbb{E}[Y_x]$ by the unbiased estimate y_x leads to *Good* and *Toulmin*'s intriguing estimator:

$$\widehat{\mathbb{E}[N]} = y_1 t - y_2 t^2 + y_3 t^3 - \dots$$

which played a significant role in their World War II code-breaking efforts

- Good credited his fellow code-breaker *Alan Turing* for some of the ideas here

- Replacing $\mathbb{E}[Y_x]$ by the unbiased estimate y_x leads to *Good* and *Toulmin*'s intriguing estimator:

$$\widehat{\mathbb{E}[N]} = y_1 t - y_2 t^2 + y_3 t^3 - \dots$$

which played a significant role in their World War II code-breaking efforts

- Good credited his fellow code-breaker *Alan Turing* for some of the ideas here
- The *Good–Toulmin* Estimator has been shown to be a good estimate for values of $t \leq 1$

- Replacing $\mathbb{E}[Y_x]$ by the unbiased estimate y_x leads to *Good* and *Toulmin*'s intriguing estimator:

$$\widehat{\mathbb{E}[N]} = y_1 t - y_2 t^2 + y_3 t^3 - \dots$$

which played a significant role in their World War II code-breaking efforts

- Good credited his fellow code-breaker *Alan Turing* for some of the ideas here
- The *Good–Toulmin* Estimator has been shown to be a good estimate for values of $t \leq 1$
- For $t > 1$, however, the Good–Toulmin estimator *fails* to capture accurate results. This is because, if $t > 1$, the Good–Toulmin estimate grows *super-linearly* in t , while the number of missing unseen species can grow at most *linearly* in t

- Replacing $\mathbb{E}[Y_x]$ by the unbiased estimate y_x leads to *Good* and *Toulmin*'s intriguing estimator:

$$\widehat{\mathbb{E}[N]} = y_1 t - y_2 t^2 + y_3 t^3 - \dots$$

which played a significant role in their World War II code-breaking efforts

- Good credited his fellow code-breaker *Alan Turing* for some of the ideas here
- The *Good–Toulmin* Estimator has been shown to be a good estimate for values of $t \leq 1$
- For $t > 1$, however, the Good–Toulmin estimator *fails* to capture accurate results. This is because, if $t > 1$, the Good–Toulmin estimate grows *super-linearly* in t , while the number of missing unseen species can grow at most *linearly* in t
- There are interesting recent work on how to improve on the Good–Toulmin estimator for $t > 1$

Species discovery curve

